POD NOVA • AUGUST COHORT • BUSINESS ANALYSIS

Checkpoint

Who decides what, and when?

Redesigning the discharge and follow-up workflow at Lakeside Health Network so humans and AI each own the decisions they're best placed to make.

TEAM

- Helena Delali Vormawor
- Phionah Atukunda, MPH, CHES®
- Angela Eze
- Olushola Omotosho

Focus Area 2: AI and the Future of Work

THE BUSINESS PROBLEM

Two hospitals merged. One AI tool. No decision owner.

Lakeside Health Network wants AI to predict which patients will be readmitted within 30 days of discharge, so nurses can prioritise follow-up. The technology is the easy part — the hard part is deciding who owns each decision once AI is in the loop.

1

Two systems, two processes

Lakeside General and Riverside Specialist ran independent discharge workflows before the merger.

2

Duplicate patient identities

The same patient can exist under two different IDs, one at each hospital.

3

No one owns the decision

Nobody has defined which AI decisions can be automated and which need a human.

OUR SOLUTION & METHODOLOGY

AI narrows the decision. A named human owns it.

Checkpoint scores and tiers patients at discharge, then routes every tier through a named human checkpoint before it reaches a patient.

1

Stakeholder analysis

Mapped everyone affected; one accountable owner per decision (RACI)

2

Process mapping

AS-IS discharge workflow at both hospitals → TO-BE workflow with AI

3

Decision-rights design

Every decision classified: automated, AI-recommends, or human-only

4

Requirements & validation

BRD, use cases, traceability, and a shadow pilot before go-live

Three jobs are redesigned around this principle: follow-up nurse, Health Records steward, ward staff.

HOW IT WORKS

AI scores and sorts. Humans decide.



Four mandatory human checkpoints

HC1

Confirm uncertain identity match

HC2

Manually assess low-confidence cases

HC3

Approve or override every tier

HC4

Monthly governance review of overrides

GOVERNANCE MODEL

Which decisions AI makes alone — and which need a person

A — Automate

Rule-based, reversible, no direct care impact

B — AI recommends, human approves

A named person must confirm first

C — Human decides

AI may inform, never choose

Decision	Level	Accountable role
Link records on an exact multi-identifier match	A	Health Records
Assign tier when data is complete and confident	B	Follow-up nurse
Assign tier when confidence is low	C	Follow-up nurse
Change thresholds or release a new model version	C	Governance committee

Full ten-decision table is in the business case document, Section 5.

OPTIONS ANALYSIS

Four options considered — one that fits the brief

Option	Why (or why not)	Chosen
A. Do nothing	High-risk patients keep getting missed; the merger's process gaps persist	No
B. Harmonise process only, no AI	Fixes some data problems, but never targets follow-up effort	No
C. AI-supported, human-in-the-loop	Targets follow-up, keeps accountability with clinicians, builds trust	Yes
D. Full automation	No safeguard against data errors or drift; highest patient-safety risk	No

Why Option C

It's the only option that uses AI to target follow-up and satisfies the brief's requirement for defined human-review points — and its costs are proportionate to the benefit case.

BUSINESS CASE AT A GLANCE

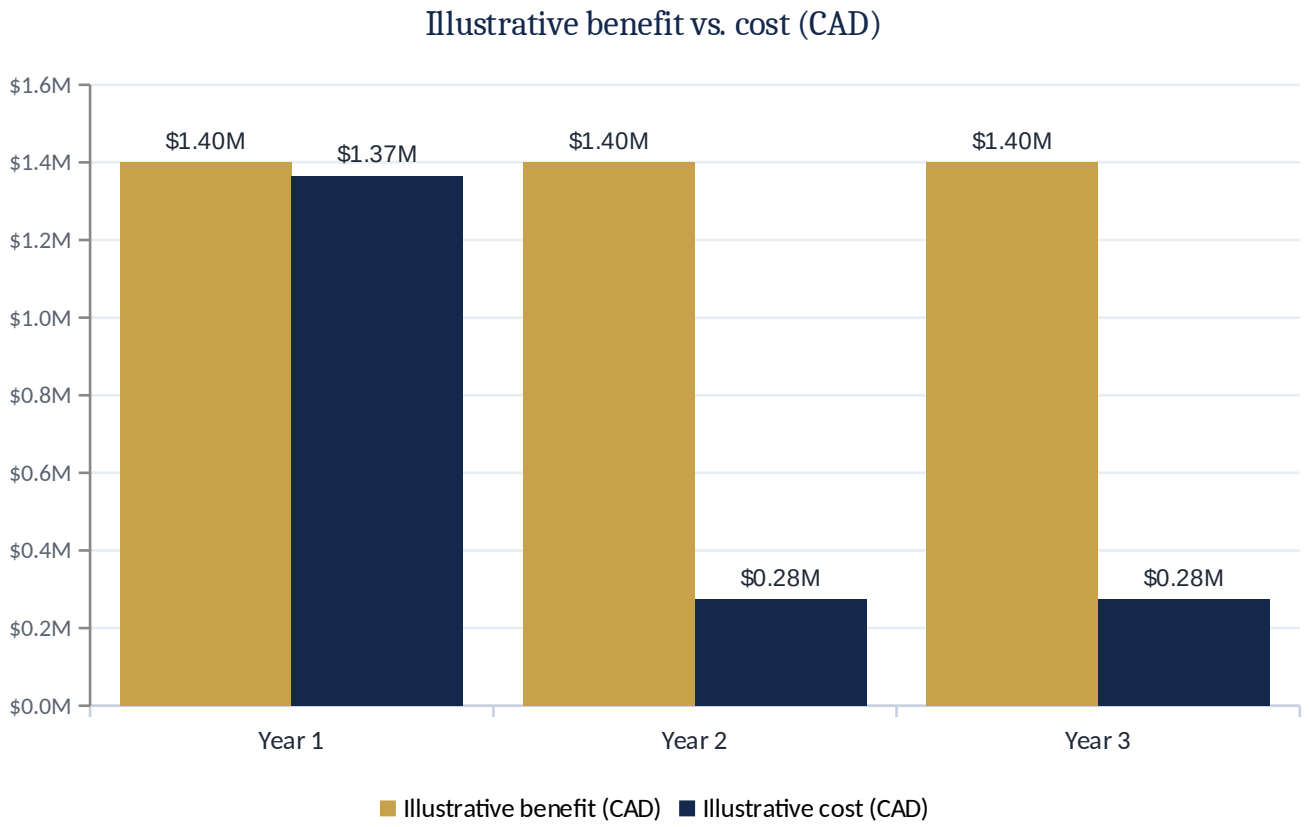
The case for Checkpoint, in one slide

Dimension	Summary
Problem	Two merged hospitals, inconsistent processes, duplicate patient identities, no decision-rights model for AI
Recommended solution	AI-scored, human-approved discharge follow-up workflow with 4 checkpoints and a 3-level automation policy
Illustrative one-time cost	CAD 1.09 million
Illustrative annual net benefit	CAD 1.18 million per year, after ongoing costs
Illustrative payback	Under 12 months
Primary risk	Automation bias and low nurse trust — mitigated by mandatory review and override rights
Timeframe	28 weeks to supervised go-live, then phased scale-up

All figures are illustrative order-of-magnitude estimates, to be recalibrated with Lakeside's own data before an investment decision. Full detail in the business case, Sections 1, 7 and 8.

FINANCIAL SUMMARY

Illustrative benefit outpaces cost from year one



Payback period

~11 to 12 months from go-live

3-year cumulative net benefit

CAD 2,285,000 (illustrative)

Simple 3-year return

~210% on the one-time investment

The readmission-avoidance benefit is the most sensitive assumption, and the first to validate in the shadow pilot.

KEY RISKS & MITIGATIONS

The risks we designed for, up front

Automation bias

Mitigation: Mandatory review (HC3), visible confidence and drivers

Wrong record merge

Mitigation: Reversible merges, HC1 steward review

Low nurse trust

Mitigation: Override rights, plain-language drivers, co-design

Uneven model performance

Mitigation: Subgroup monitoring, monthly governance review (HC4)

IMPLEMENTATION ROADMAP

28 weeks to a supervised go-live



Prepare	Clean up & train	Shadow pilot	Supervised go-live	Scale & embed
Wks 1–4	**Wks 5–12**	**Wks 13–20**	**Wks 21–28**	**Wk 29+**
Baseline, privacy assessment	Fix duplicates, train staff	Score without acting, validate	One ward per site	Extend network-wide

Shadow pilot (Phase 2) is the key gate: it validates the benefit case before wider investment.

SUCCESS CRITERIA & KPIS

How we'll know Checkpoint is working

Every measure is SMART — a specific number, a named baseline, a fixed review cycle. Figures are illustrative targets pending the Phase 0 baseline.

Measure	Illustrative target	Review
High-tier patients contacted	≥90% within 2 working days of discharge	Monthly
30-day readmission, High-tier	≥15% reduction vs. baseline within 6 months	Quarterly
Override rate vs. AI tier	10–20% band; any reason code >10% triggers review	Monthly, HC4
Duplicate record rate	From ~8% baseline to <2% within 90 days	Monthly, Phase 1
Nurse trust score	≥80% rate the tool 4/5 or higher	End of pilot
Subgroup performance gap	≤5 percentage points across subgroups	Monthly, HC4

ALIGNMENT TO THE CHALLENGE

Built for Focus Area 2: AI and the Future of Work

> *"How can organisations redesign jobs and workflows to help humans work effectively alongside AI?"*

Follow-up nurse

From building call lists by hand, to reviewing a prioritised worklist and owning the decision

3 **Good Health & Well-Being**

Better-targeted follow-up

Health Records team

From filing and retrieval, to network patient-identity stewardship

10 **Reduced Inequalities**

One standard of care, both sites

Ward staff

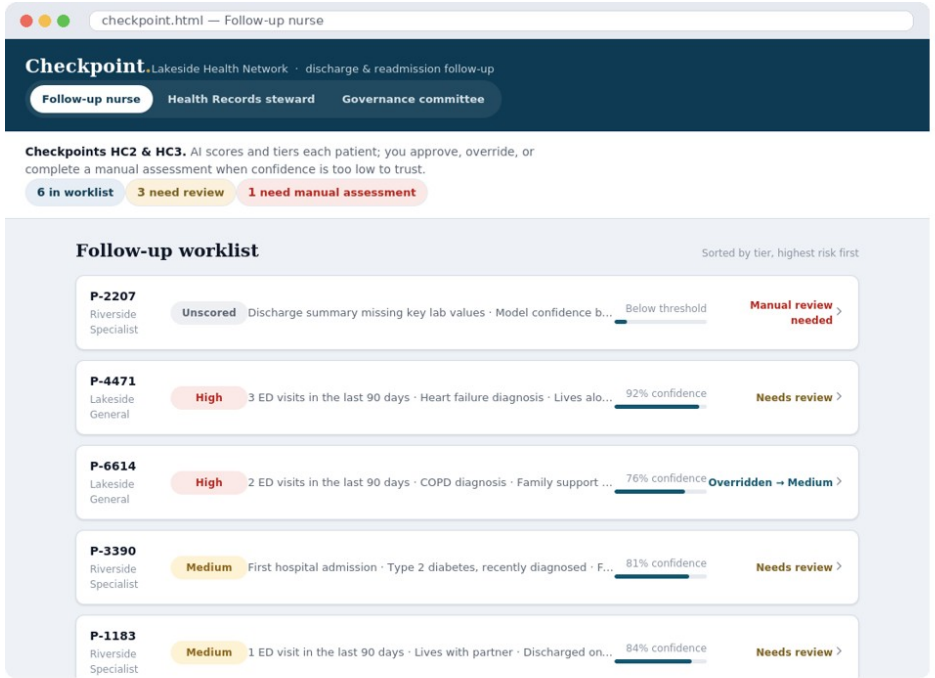
From two discharge standards, to one harmonised process

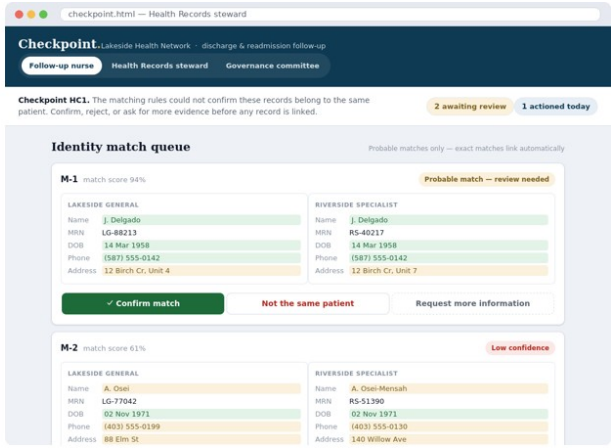
17 **Partnerships for the Goals**

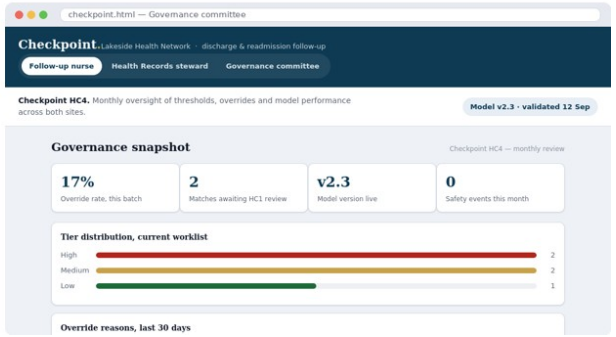
Governed data sharing

WORKING PROTOTYPE

Checkpoint, built and clickable







Three roles, one tool

- Follow-up nurse: approve or override every tier
- Health Records: resolve identity matches
- Governance: monthly oversight of overrides
- Every action logged and reversible

LIVE DEMO

lenavon.github.io/10alytics-Hackathon---Checkpoint-/

A recorded demo is included with this submission

RECOMMENDATION & THANK YOU

Proceed with a shadow pilot

Prove the decision model and the benefit case before wider investment.

1	**Secure sponsorship**	Board of Directors sponsor and a governance committee with clear terms of reference
2	**Baseline the numbers**	Replace illustrative figures with Lakeside's real duplicate rate, readmissions and follow-up timing
3	**Run the shadow pilot**	20 weeks to a validated benefit case and a go or no-go decision, before scaling

Thank you. Checkpoint — a human-centred path to AI-supported care at Lakeside Health Network.

Developed with the assistance of Claude (Anthropic) for drafting, diagrams and prototype code. All analysis, requirements decisions and business judgement are the work of the Pod Nova team.